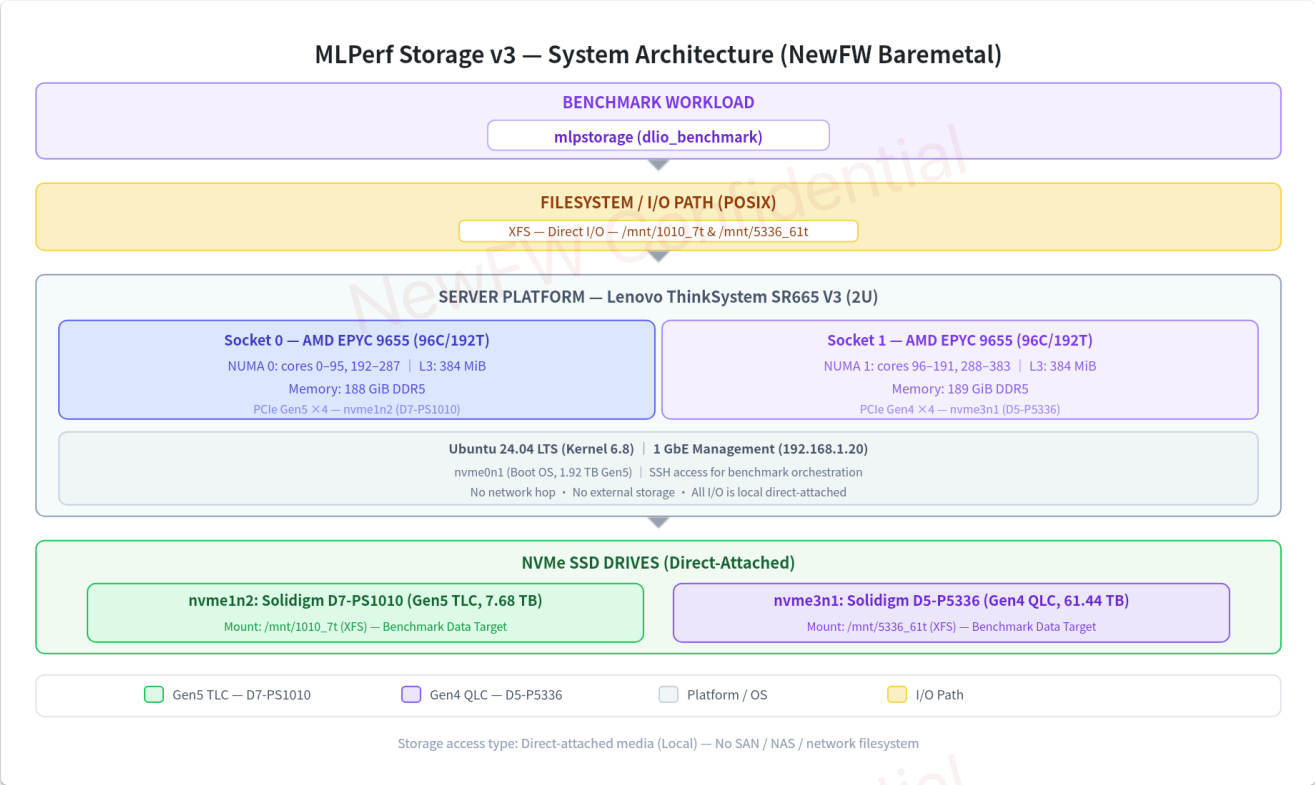


1. Overview

Deployment: 1x Lenovo ThinkSystem SR665 V3 (2U), Ubuntu 24.04 LTS.

Test methodology: Single-drive, single-system. Two Solidigm NVMe SSDs tested independently — D7-PS1010 (Gen5 TLC, 7.68 TB) and D5-P5336 (Gen4 QLC, 61.44 TB) — each as a local XFS mount. All I/O is direct-attached; no SAN, NAS, or network filesystem.

2. System Architecture



Storage Data Path (Local): mlpstorage → POSIX read/write (XFS) → /mnt/1010_7t or /mnt/5336_61t → NVMe Block Device. No network hop, no external storage, all I/O is local.

3. Hardware Components

3.1 Server Platform

Component	Specification
Model	Lenovo ThinkSystem SR665 V3
Form Factor	2U rack-mount
Chassis	Standard Lenovo 2U

3.2 Processors

Component	Specification
CPU Model	AMD EPYC 9655 96-Core Processor
Sockets	2
Cores per Socket	96
Total Physical Cores	192
Threads (SMT)	384
Base Frequency	2.6 GHz
L1d Cache	9 MiB (192 instances)
L1i Cache	6 MiB (192 instances)
L2 Cache	192 MiB (192 instances)
L3 Cache	768 MiB (24 instances)
NUMA Nodes	2

3.3 Memory

Component	Specification
Total Capacity	377 GiB (~395,565,104 kB as reported by /proc/meminfo)
Type	DDR5
Configuration	12 × 32 GB
Distribution	~188 GiB per NUMA node

3.4 Storage Devices

Benchmark Data Drives

Attribute	Drive 1 (TLC)	Drive 2 (QLC)
Device	/dev/nvme1n2	/dev/nvme3n1
Model	Solidigm D7-PS1010	Solidigm D5-P5336
Part Number	SB5PH27X076T	SBFPF2BV614T
Capacity	7.68 TB	61.44 TB
NAND Type	TLC 3D	QLC 3D
Interface	PCIe 5.0 ×4 NVMe	PCIe 4.0 ×4 NVMe
Firmware	G70YG250	5CV10202
Mount Point	/mnt/1010_7t	/mnt/5336_61t
Filesystem	XFS	XFS
Role	Primary benchmark target	Secondary benchmark target

Boot/OS Drive

Attribute	Value
Device	/dev/nvme0n1
Model	Solidigm SB5PH27X019TOP
Capacity	1.92 TB
Interface	PCIe 5.0 x4 NVMe
Role	Operating system + benchmark software

Note: Per MLPerf Storage rules, the benchmark results directory (`--results-dir`) is placed on the OS drive (`nvme0n1`), which is a **different filesystem** from the data drive (`/mnt/1010_7t` or `/mnt/5336_61t`). This ensures log I/O does not interfere with storage benchmark measurements.

3.5 Networking

Interface	Speed	IP Address	Role
Broadcom BCM5720 (enxc8a...)	1 GbE	192.168.1.20	Management / SSH
USB Ethernet (enx3a7c...)	1 GbE	192.168.1.30	Secondary

Since this is a **single-host** submission using local storage only, networking plays no role in the storage data path. Both interfaces are used for remote administration only.

3.6 Power

Component	Specification
PSU Configuration	1+1 Redundant (Active/Passive)
PSU Rating	1800W each
Design Power	1800W
Active PSUs	1
Passive PSUs	1

Power consumption was not directly measured during benchmarks. The design power is based on the PSU nameplate rating.

4. Software Stack

4.1 Operating System

Component	Version
Distribution	Ubuntu 24.04 LTS (Noble Numbat)
Kernel	6.8.0-31-generic

Component	Version
Architecture	x86_64

4.2 Benchmark Software

Component	Version / Source
MLPerf Storage (mlpstorage)	v3.0.42, commit 19613128 ; VDB: v3.0.44, commit 036406a0
Repository	https://github.com/mlcommons/storage
DLIO Benchmark	Bundled via uv (pyproject.toml lockfile)
Python	3.12.3
Package Manager	uv (installed at /root/.local/bin/uv)
MPI	Open MPI 4.1.6
Workspace Path	/root/raysmond/workspace/mlperf_v3/storage

4.3 Software Modifications

None. The benchmark code is unmodified from the official `mlcommons/storage` main branch. No patches have been applied to the repository source or to the installed DLIO dependencies.

5. System Tuning

5.1 NewFW-Tuned Configurations

Beyond the explicit parameters above, the following subsystems have been carefully tuned by NewFW for optimal storage benchmark performance:

- Kernel VFS parameters:** Virtual filesystem tunables (e.g., dirty ratios, readahead, scheduler settings) are adjusted from Ubuntu defaults to maximize NVMe throughput and reduce I/O latency under benchmark workloads.
- NVMe formatting:** NVMe namespaces are formatted with optimal LBAF (Logical Block Address Format) and namespace geometry aligned to the benchmark I/O patterns.

```
mkfs.xfs -f -K -d agcount=32 /dev/nvmeXnY
```

- Filesystem creation & mount:** XFS filesystems are created with allocation groups (`agcount`), stripe alignment (`sunit` , `swidth`), and other `mkfs.xfs` parameters tuned to the underlying NVMe device characteristics; mount options (e.g., `noatime`) are selected to minimize metadata overhead.

`/etc/fstab` entries:

```
UUID=5970b440-445a-4c75-9e96-649f551db7db /mnt/1010_7t xfs
rw,nofail,noatime,nodiratime,attr2,largeio,inode64,logbufs=8,logbsize=256k,noquot
a 0 0
UUID=56ea2fe2-cd83-4ba4-8ae9-c7def5bf6973 /mnt/5336_61t xfs
```

```
rw,nofail,noatime,nodiratime,attr2,largeio,inode64,logbufs=8,logbsize=256k,sunit=32,swidth=32,noquota 0 0
```

All other general OS kernel parameters, sysctl values, and system configurations remain at Ubuntu 24.04 LTS defaults. No GPU hardware is present or used.

6. Submission Configuration

6.1 Single-Host Declaration

This is a **single-host submission** as defined in the MLPerf Storage rules:

- 1 client host runs all benchmark processes
- Data resides on locally-attached NVMe SSDs
- No external storage system, no network storage fabric
- Storage access type: **Direct-attached media (Local)**

7. Additional Information

7.1 Accessibility

All benchmark tests are dispatched from the NewFW CWP (Cloud Workload Platform) to the local system via the `cwp-agent` command-line tool, which executes MLPerf Storage test scripts and collects results. CWP (<https://cwp.newfw.com.cn>) is a cloud workload platform for workload management, distribution and scheduling, execution, monitoring, and result data/log collection.

7.2 Contact

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